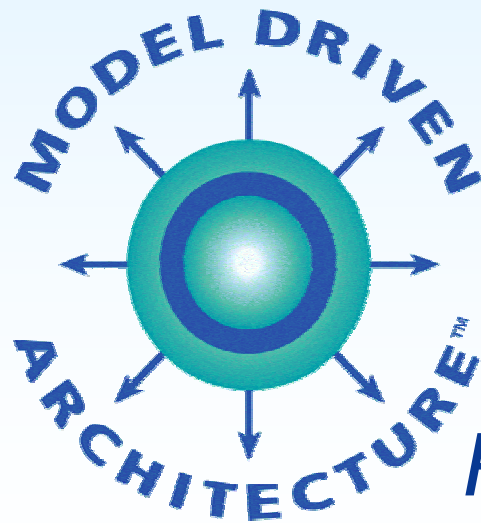
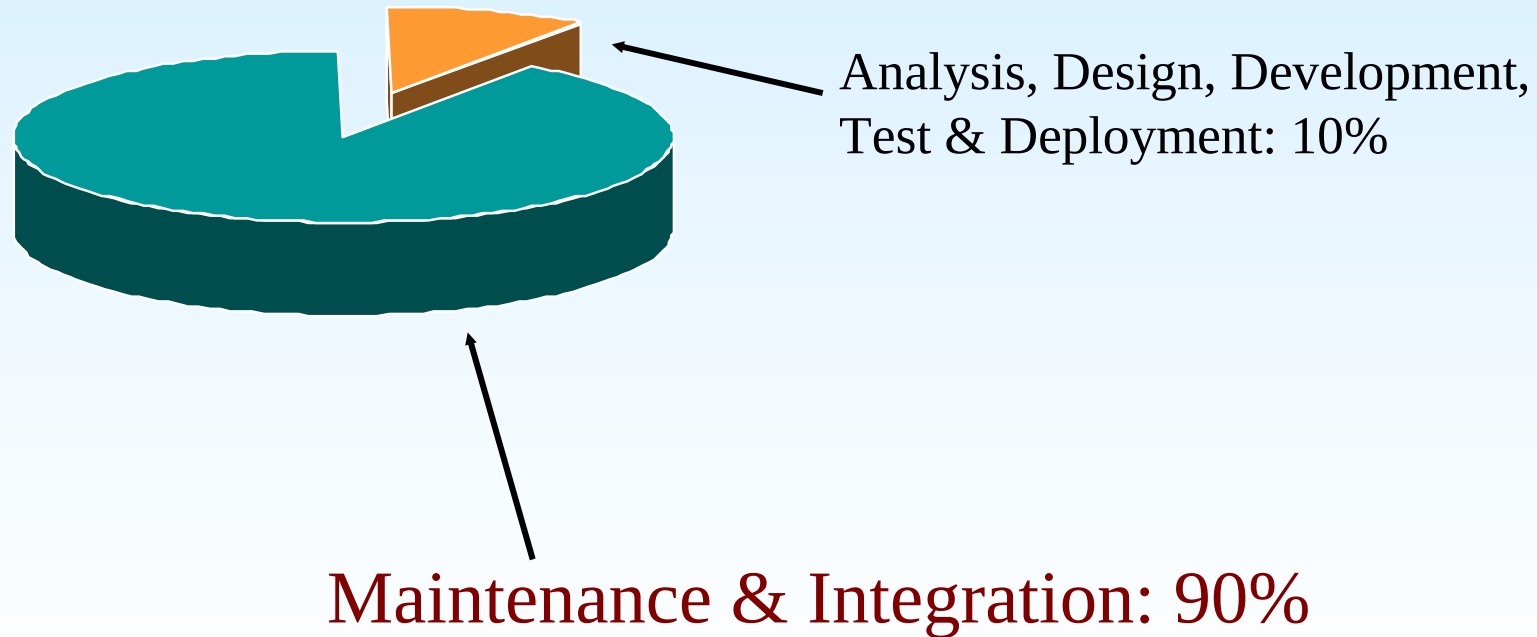


Model Driven Architecture: Three Years On



*Richard Mark Soley, Ph.D.
Chairman and CEO*

Are You Spending Wisely?



Roadkill on the Info Highway



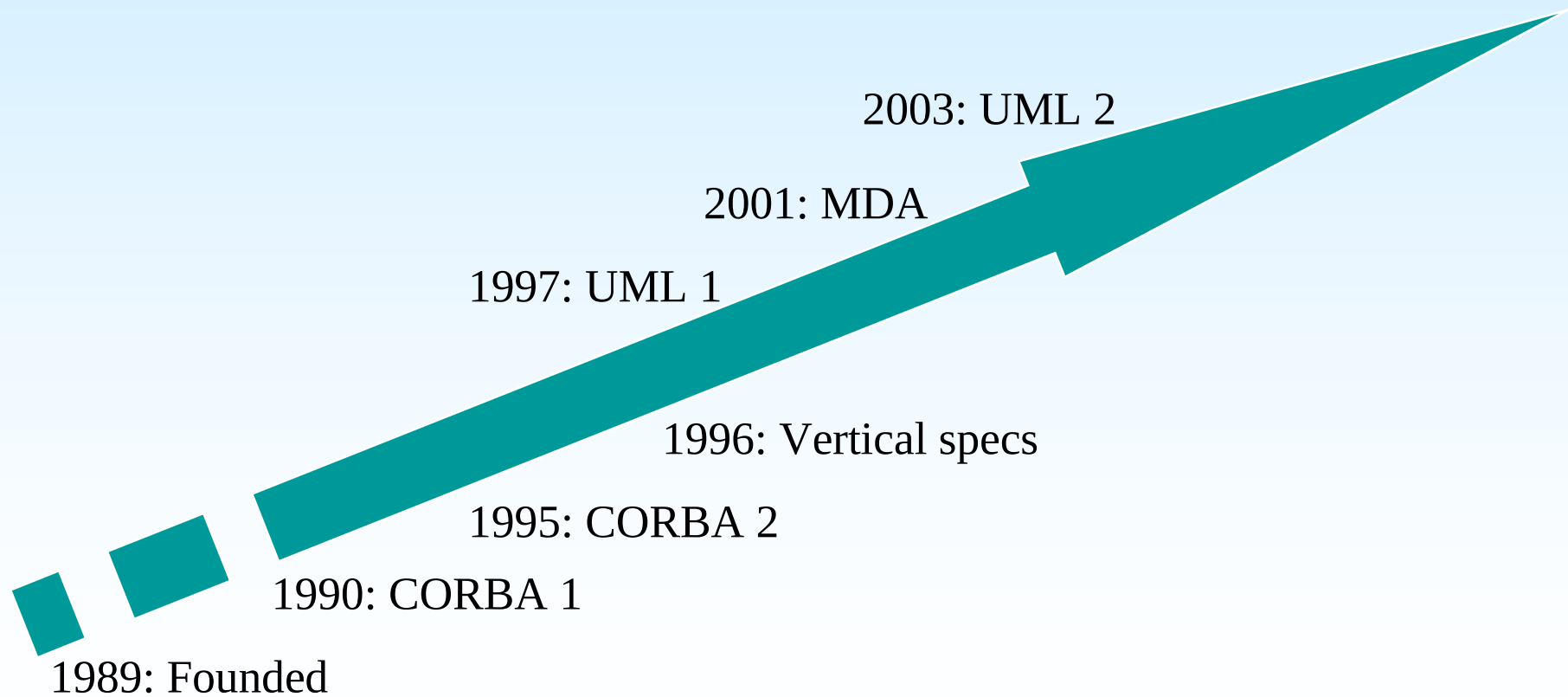
Doing things “the way we always do them” isn’t the answer.

Model Driven Architecture

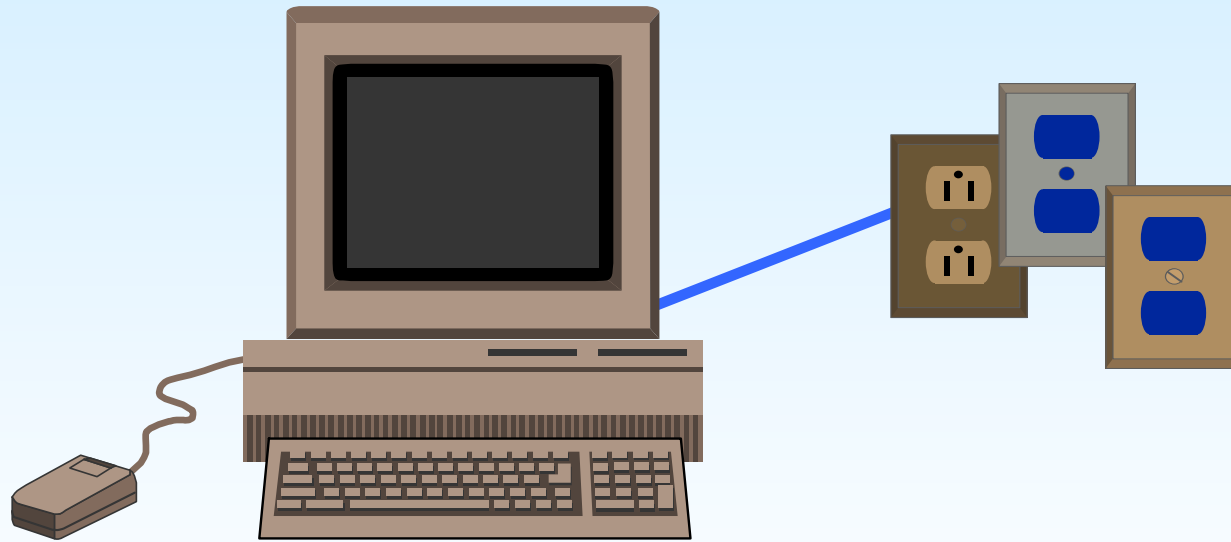
- An *initiative* of the Object Management Group (OMG)
- A *brand* for tools based on OMG's UML and MOF open standards
- A *set of specifications* defined by OMG's open worldwide process

Initiated late 2000; public since 2001.

OMG: An Interesting History

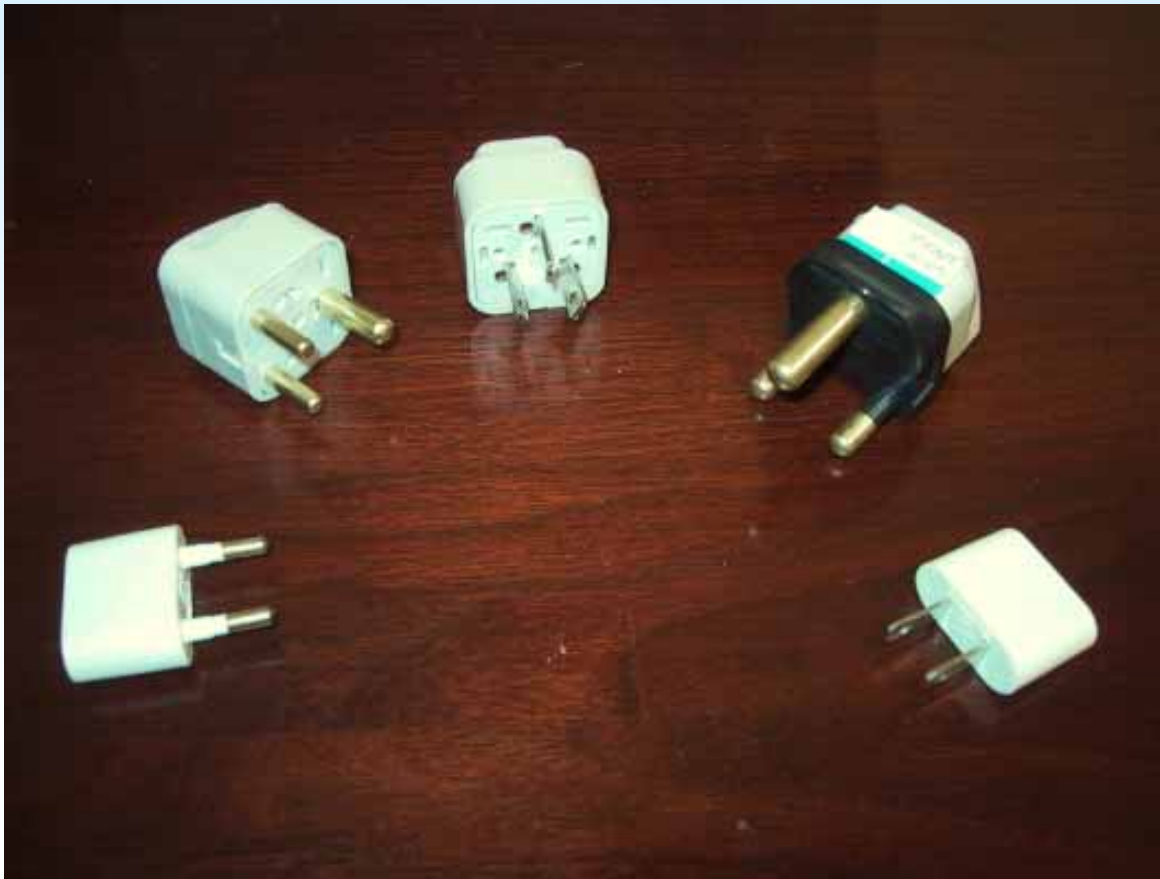


OMG's Vision



The Global Information Appliance

Not too bad for electrical
power



...but a mess for telephony!



Heterogeneity is Permanent

- Programming languages
 - ~3 million COBOL programmers
 - ~1.6 million VB programmers
 - ~1.1 million C/C++ programmers
- Operating systems
 - Unix, MVS, VMS, MacOS, Windows, PalmOS, Symbian...
 - Windows 3.1: it's still out there!
 - Embedded devices (mobile, set-top, etc.)
- Networks
 - Ethernet, ATM, IP, SS7, Firewire, USB
 - Bluetooth, 802.11b, HomeRF

Where Can We Agree?

- There will not be consensus on hardware platforms
- There will not be consensus on operating systems
- There will not be consensus on network protocols
- There will not be consensus on programming languages
- ***There must be consensus on models, interfaces and interoperability!***



OMG's Mission Since 1989

- Develop an architecture, using object technology, for distributed application integration, guaranteeing:
 - reusability of components
 - interoperability & portability
 - basis in commercially available software
- Specifications *freely available*
- Implementations exist
- Member-controlled not-for-profit

Who Are OMG?



The following table lists the companies shown on the map, organized by their geographical location as indicated by the map:

North America	Europe	Asia	Oceania
Alcatel	Fujitsu	John Deere	PRISM Technologies
AT&T	Glaxo SmithKline	LION Bioscience	Pfizer
BEA Systems	Hewlett Packard	Lockheed Martin	SAP
Borland	Hitachi	MITRE	SAS Institute
Boeing	Hyperion	Motorola	Softteam
CA	IBM	NASA	Sun Microsystems
Codagen	IONA	NEC	TCS
Compuware	io Software	Nokia	Unisys
DaimlerChrysler	Kabira	NTT DoCoMo	VHA
EDS	Kennedy Carter	OASIS	W3C



How Can We Protect Software Investment?

- The problem is
 - Tracking the *next best thing*
 - Protecting your investment in existing software base
 - Retaining qualified staff
 - Maintaining existing code base
- Integrating what you've built
 - With what you are building
 - *With what you will build!*

The Model Driven Architecture

- OMG's *Model Driven Architecture* (MDA™) initiative is aimed precisely at this problem
- You have an opportunity to increase your bottom line by *integrating your assets*
- Industry standards support that goal by future-proofing your application design
- The MDA will help you integrate the mix you have today, and give you an architecture to support the unexpected
- Focus on integrating legacy applications
- Ensure smooth integration of COTS applications
- Models are *testable* and *simulatable*
- The aim: *a 20-year software architecture*

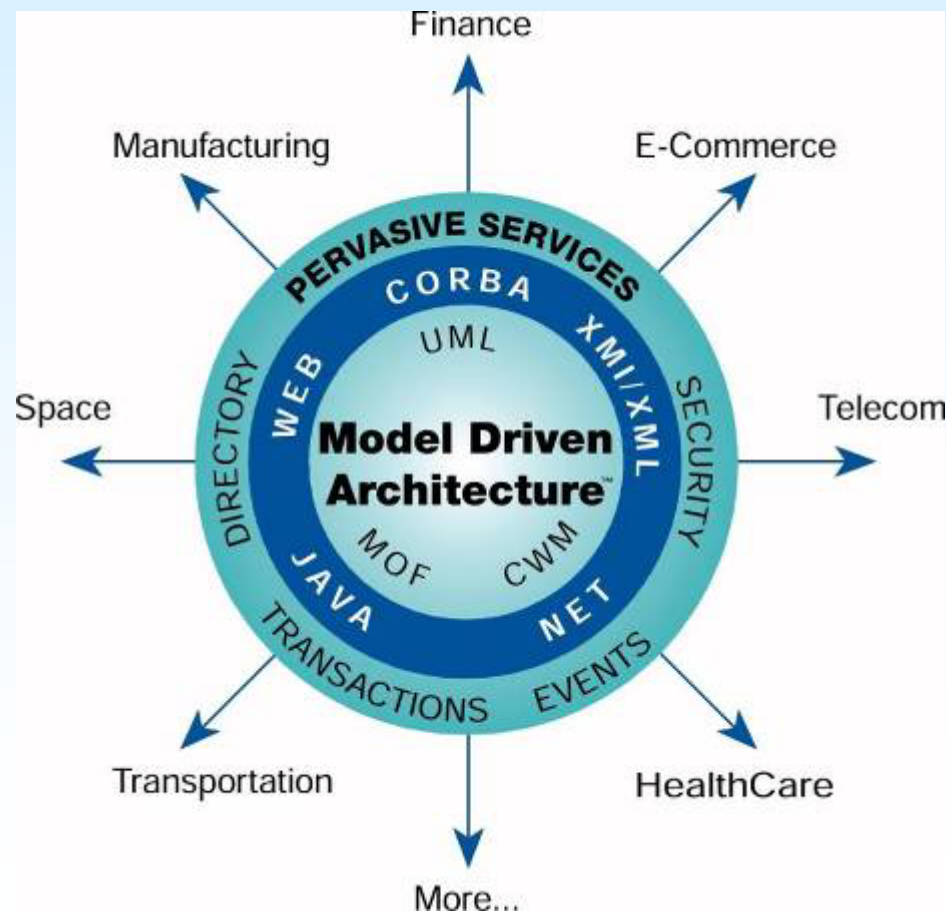
What is Model Driven Architecture?

- A New Way to Specify and Build Systems
 - ***Based on modeling with UML***
 - Supports full lifecycle: analysis, design, implementation, deployment, maintenance, evolution & integration with later systems
 - Builds in Interoperability and Portability
 - Lowers initial cost and maximizes ROI
 - Applies directly to the mix you face:
 - Programming language
 - Operating system
 - Network
 - Middleware

OMG's Modeling Suite

- Unified Modeling Language
 - UML™ remains the world's only standardized object-oriented modeling language, best-known part of the standard
- Common Warehouse Metamodel
 - CWM™, the integration of the last two data warehousing initiatives
- Meta-Object Facility
 - MOF™, the metadata integration standard
 - Defines metadata & metadata services
- XML Metadata Interchange
 - XMI™, the XML-UML standard
 - Cornerstone of UML tool interoperability
 - Actually just a collection of mapping rules XML/MOF

Model Driven Architecture



Leveraging UML is Critical

- The Unified Modeling Language is the successor to the dozens of OO A&D notations of the early '90s.
- Result of an OMG adoption begun in '96 and completed in '97
- Complemented with repository (MOF) and XML Metadata specs (XMI)
- Standardization primed the market
 - Over 200 books, in many languages
 - Dozens of commercial tools
 - Widely available training
- Supported by an open process
 - UML 2.0 & MOF 2.0 developed openly

MDA Books Proliferating



UML Certification Program



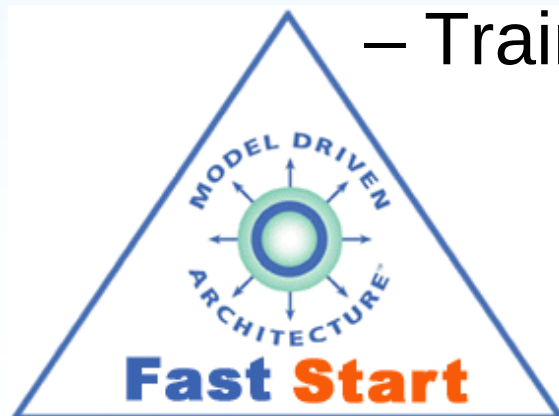
OMG/UTI UML Certification Program

- Starts November 2003
- Worldwide (English & Japanese)
- Three levels of accomplishment
- Focused on UML 2.0
- **OMG ENDORSED**

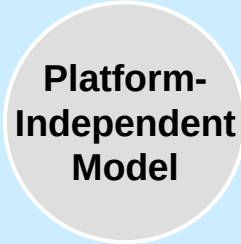
MDA FastStart

Fast introduction to MDA technologies

- Initial project selection and planning
- Consulting
- Mentoring
- Training



Building an MDA Application

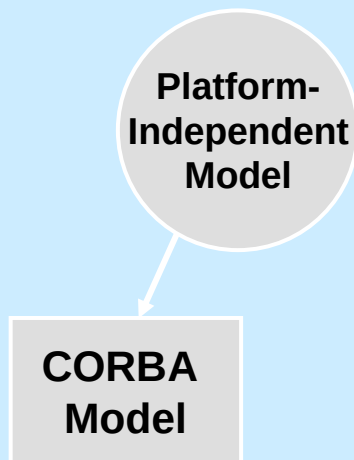


Platform-
Independent
Model

A Detailed Model,
stating Pre- and Post-
Conditions in OCL,
and Semantics in
Action Language

Start with a *Platform-Independent Model* (PIM) representing business functionality and behavior, undistorted by technology details.

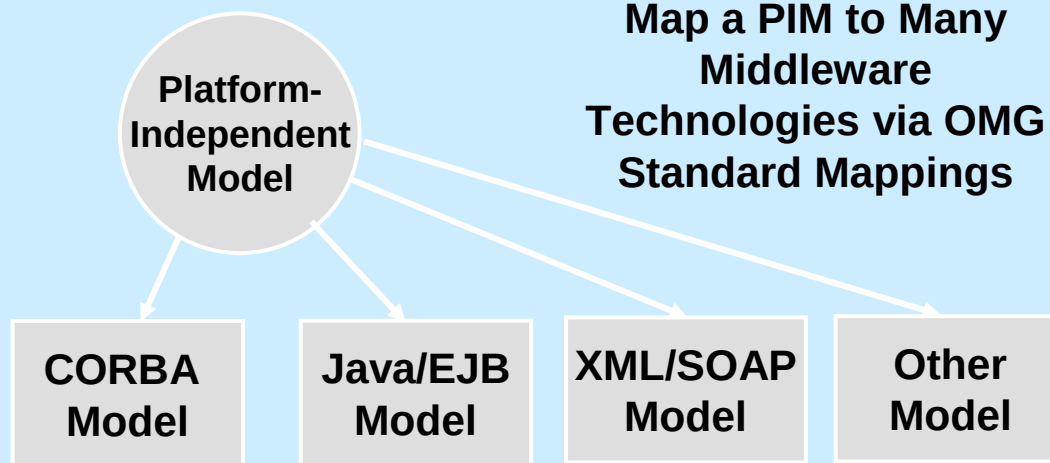
Generating Platform-Specific Model



Map a PIM to Specific
Middleware
Technologies via OMG
Standard Mappings

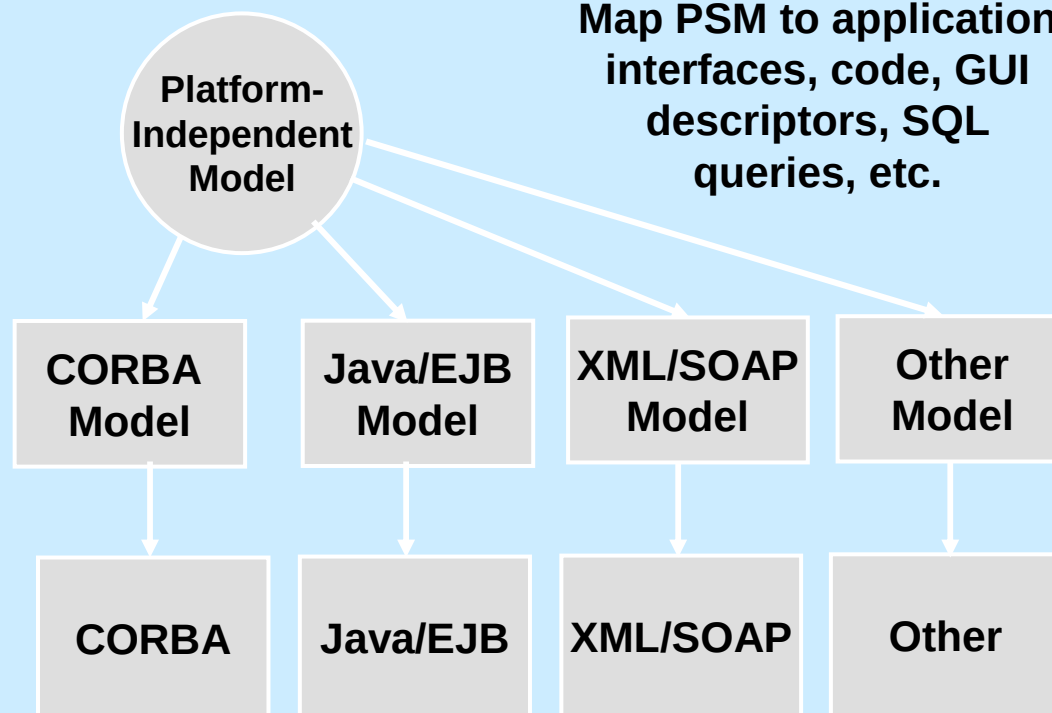
MDA tool applies a standard mapping to generate *Platform-Specific Model* (PSM) from the PIM. Code is partially automatic, partially hand-written.

Mapping to Multiple Deployment Technologies



MDA tool applies an standard mapping to generate *Platform-Specific Model* (PSM) from the PIM. Code is partially automatic, partially hand-written.

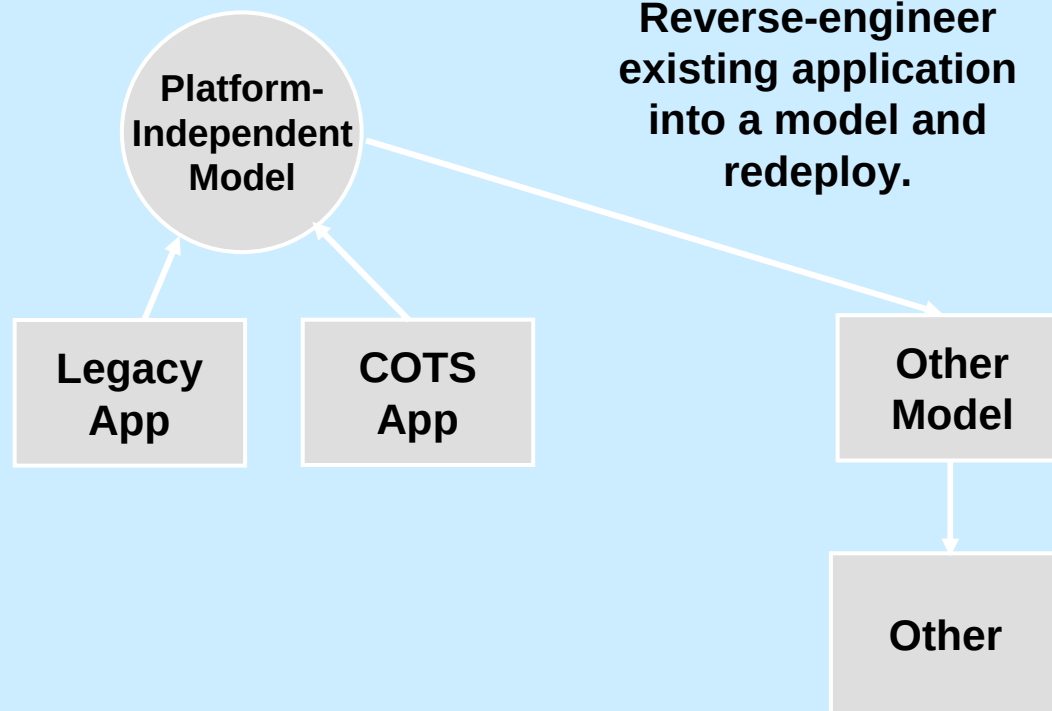
Generating Implementations



Map PSM to application interfaces, code, GUI descriptors, SQL queries, etc.

MDA Tool generates all or most of the implementation code for deployment technology selected by the developer.

Integrating Legacy & COTS



Reverse-engineer existing application into a model and redeploy.

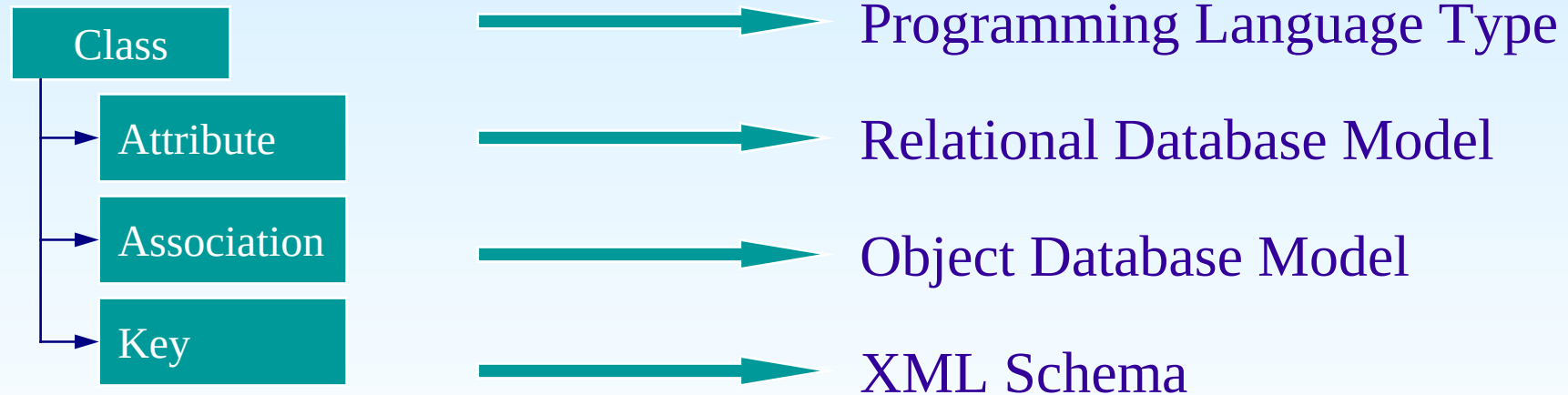
MDA Tools for reverse engineering automate discovery of models for re-integration on new platforms.

Data Integration Works Too

- MOF is the key here, behind the UML scene
- With one modeling language
 - Anything can be modeled
 - User must map (code) from domain to modeling language constructs
- MOF enables definition of multiple modeling languages
 - Specific to various domains
 - Semantics captured in metamodels
 - User no longer needs to “map”, tools do

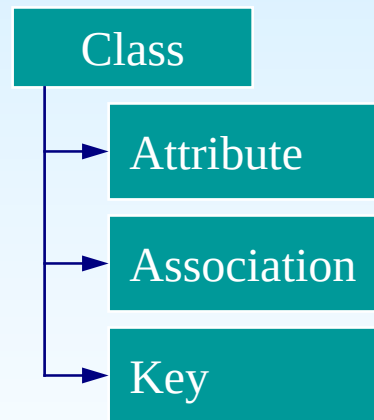
Data Integration: Choices

Meta-meta-model

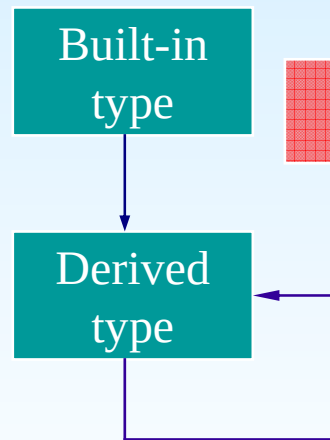


Data Integration: Data Model

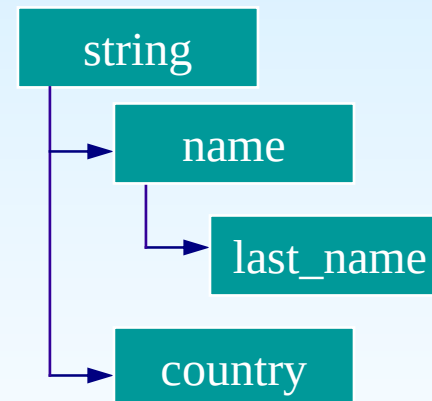
Meta-meta-model



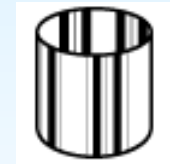
Meta-model



Model

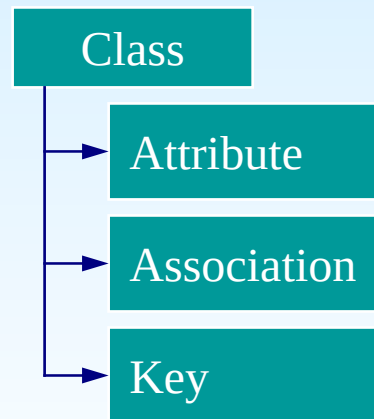


Data

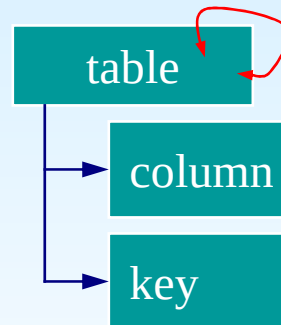


Data Integration: Relational

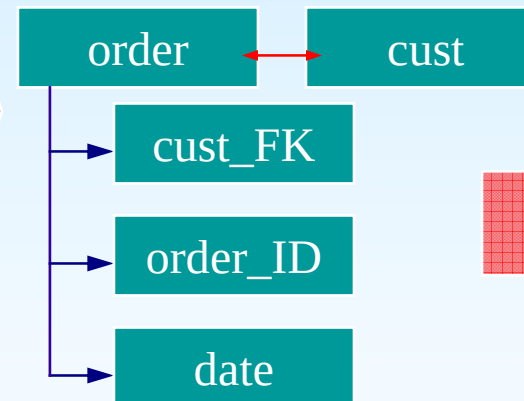
Meta-meta-model



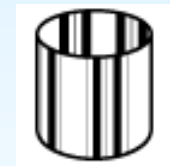
Meta-model



Model

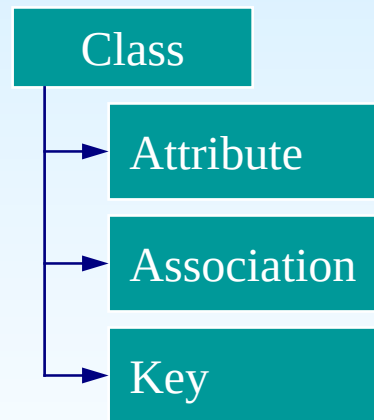


Data

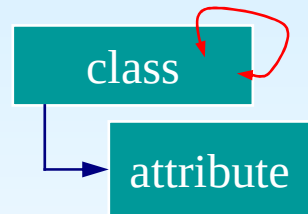


Data Integration: Object

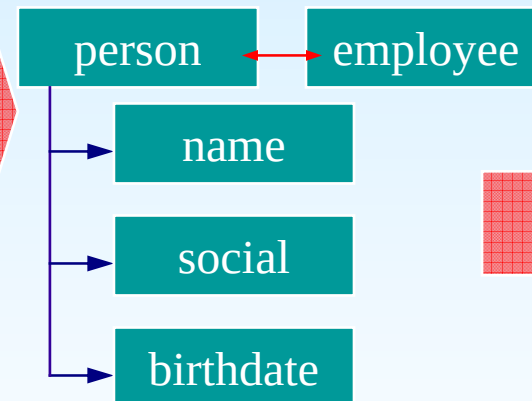
Meta-meta-model



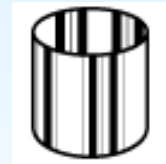
Meta-model



Model

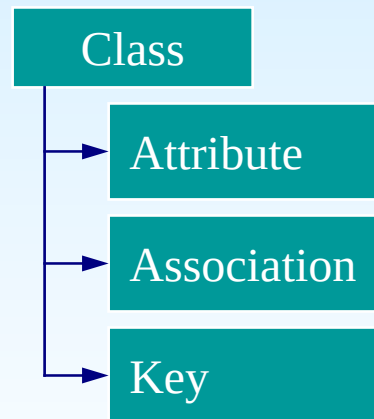


Data

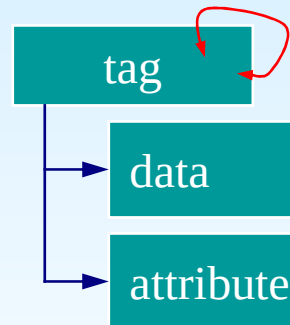


Data Integration: XML

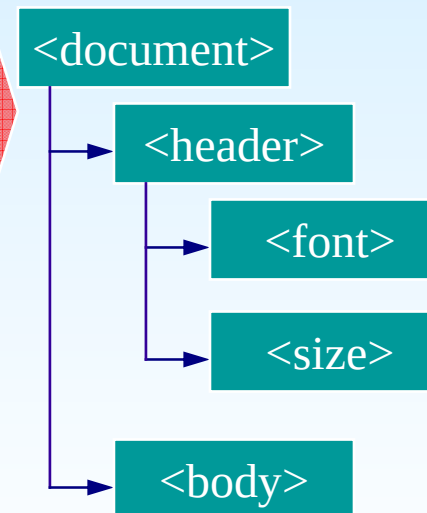
Meta-meta-model



Meta-model



Model



Data



MDA in Industry Standards

- The MDA promotes standards that are valuable *across deployment technologies*
 - Applicable to large & small deployments, new applications, legacy and COTS
 - Applicable to CORBA, DCOM, .Net, etc.
 - Allows knowledge leverage for the long-term, standards that persist
- MDA has been quickly adopted by OMG's standardization groups
 - Both PIM and PSM(s) adopted by process
 - Standard model lasts decades

OMG MDA Adoption Status

- Major direction agreed March '01; overall architecture adopted September '01.
- UML 2.0 completed in '03
- Large suite of standards already done:
 - Core: UML, MOF, CWM, XMI
 - Enterprise Models: EDOC, EAI
 - Platforms: RT, CORBA, J2EE
 - Platform soon: WS, .NET

OMG MDA Adoption Status

- More importantly, *vertical market groups* are thriving on MDA approach:
 - Electronic Commerce
 - Financial Services
 - Healthcare
 - Life Sciences Research
 - Manufacturing
 - Space & Ground Systems
 - Telecommunications
 - Legacy Integration

MDA Benefits

- Full support for your “20 year architecture” across the application lifecycle
- Smooth integration across intra- and inter-business boundaries (across deployment technologies)
- Reduced costs from beginning to end
- Reuse of applications, code, training and people
- Technology-independent representation of the *business*
- Scalability, robustness & security via generated code
- Stable model-based approach maximizes ROI
- Rapid inclusion of the *next best thing*

The CIO Problem Solver

MDA in Practice

- Several excellent proofs-of-concept:
 - Wells Fargo (an architecture that has already been resilient through a decade of change)
 - Lockheed Martin Aeronautics
 - GCPR in US government
 - Deutsche Bank Bauspar
 - Defense Information Systems
 - Merrill Lynch
 - Österreichische Bundesbahn
 - Thales Training & Simulation
 - Zuercher Kantonal Bank
 - CGI
 - Chubb and Son

CGI (Large Lottery Project)

- No savings in requirements, analysis or design
- Estimated total project savings >25%
- Code generation >50%



National Service Industries

- Upgrade 30-year-old legacy
- Initial proposal >5 years, >US\$5MM
- Delivered in <1 year, US\$1MM
- *Initial* 80% savings



Austrian Railways

- Immense J2EE project (400k EJB's)
- Nationwide, realtime billing system
- Initial cost savings 35%



Lockheed Martin F-16

- Modular Mission Computer
- Cross-platform mission management
- Huge maintenance & integration savings
- 100% code generation



MDA Three Years On

- 2000: initial impetus for initiative
- 2001: members carry initiative forward
- 2002: initial profiles
- 2003: UML 2, MOF 2
- 2004: BPM, Embedded



To Get More Information

- MDA Information Page
 - <http://www.omg.org/mda/>
- OMG General Information
 - <http://www.omg.org/>
- Contact the Author
 - soley@omg.org

<http://www.omg.org/~soley/mda3yearson.ppt>